



Clinical
Machine
Learning
& Hebrew
NLP

Early Prediction of Surgical-Site Infection

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01 Introduction

Surgical-site infection follows 2 to 5% of operations, yet it is still caught only once clinically apparent, days after the wound closes.

We score every operation the moment it ends, reading the structured record and the Hebrew clinical narrative together.

RESEARCH QUESTION

To what extent can a multimodal machine learning framework, combining structured tabular EHR data with unstructured narrative clinical text, predict the risk of postoperative infection at the end of the surgical procedure among surgical patients at Assuta Ramat HaHayal, compared with a single-modality tabular model?

02 Data & Methods

35,925 operations	141 infections · 0.39%	9,555 frozen test · 42
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NLP FINAL MODEL
Hebrew TF-IDF · logistic regression · average precision 0.519

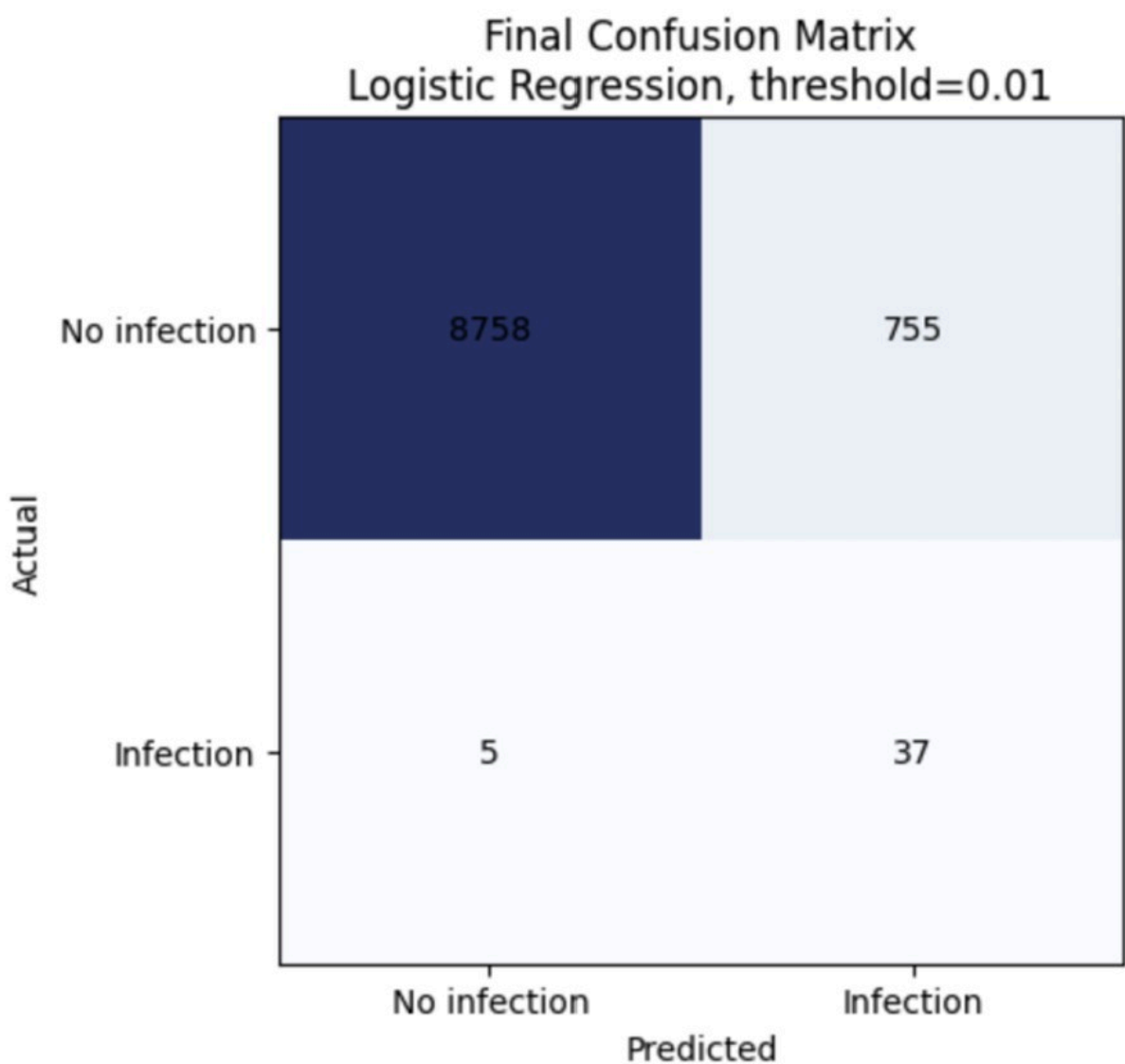
TABULAR FINAL MODEL
Logistic regression · SMOTE · isotonic calibration · threshold 0.010

03 Results Frozen test set · 9,555 operations · 42 infections

88.1% Recall	0.943 ROC-AUC	0.183 PR-AUC	21.4 Needed to screen
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The final tabular model catches 37 of 42 infections at 83 alerts per 1,000 operations.

CONFUSION MATRIX · TABULAR FINAL MODEL



TABULAR MODELS · PR-AUC

Random forest	0.259
CatBoost	0.226
Logistic regression	0.183
XGBoost	0.163

Logistic regression was selected for its far higher sensitivity to infection at comparable PR-AUC.

04 Conclusions

- The Hebrew clinical narrative carries genuine infection signal, and the simplest representation extracted the most of it.
- At this prevalence, high discrimination does not translate into high precision. Precision and alert burden describe clinical value, and the threshold is where the trade-off is made.
- The prediction point is a clinical decision, not a technical parameter: scoring was moved back to the end of surgery.

05 Discussion

- Integrate the narrative more richly than a single scalar score.
- Compare TF-IDF with a Hebrew clinical transformer fine-tuned on local text.
- Evaluate the model prospectively in surgical departments before clinical use.



CLINICAL PREDICTION MODELING